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Zr alloying effects on the microstructure, compression performance and oxidation resistance of refractory high entropy alloys

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Abstract

Three refractory high entropy alloys (RHEAs) with nominal compositions of $Mo_{0.5}NbHf_{0.25}CrZr_xTiAl$ (x = 0, 0.5, 1, mole ratio) were prepared by vacuum non-consumable arc melting technology. The influences of Zr addition on the microstructure, compressive performance and oxidation resistance at 1473 K of RHEAs were investigated and evaluated integrally. The results show that the three RHEAs are all composed of BCC1, BCC2 and Laves phases. With increasing the Zr content, the microstructure of RHEA has gradually become BCC1 + BCC2 eutectics, and also the area fraction of BCC2 phase exhibits a rising trend. Both peak stresses are displayed at the strains of near 0.03 and 0.08 respectively in the compressive stress-strain curves of the three RHEAs. On the whole, the compressive yield strength of RHEA shows an increasing trend with Zr addition. The oxide scales of the three RHEAs all possess good adhesion and obviously layered structures. The different oxidation products are distributed in their oxide scales and internal oxidation zones. The oxidation resistance of RHEA is dramatically degraded due to the decreased compactness of the oxide scale with Zr addition.

1. Introduction

The development of aerospace technology has put forward higher requirements for its high temperature hot end component materials, that is, moderate density, high temperature bearing capacity, excellent mechanical properties at room and high temperatures (toughness, ductility, strength, creep resistance), and good oxidation resistance at high temperature [1]. In recent years, refractory high entropy alloys (RHEAs) with equal or nearly equal atomic ratio, as a potential high temperature structural material, have attracted extensive attention from scholars at home and abroad because of their high temperature resistance and outstanding mechanical properties at room and high temperatures [2].

W, Ta, Mo, Nb and Hf with high melting temperatures are important constituent elements of RHEAs, which can provide excellent mechanical properties at room and high temperatures [3]. Nevertheless, the density of W and Ta elements is very high (19.35 and 16.6 g cm^{-3} respectively), and the former element is unfavorable to the high temperature oxidation resistance of RHEAs due to the formation of volatile tungsten trioxide (WO_3) [4, 5]. Mo and Nb elements have moderate density (10.2 and 8.57 g cm^{-3} respectively) and Hf element can ameliorate oxidation resistance of some high melting point alloys by improving the adhesion and compactness of oxide scale, promoting selective oxidation of Ti and other elements and inhibiting inward diffusion of oxygen [6–8].

Chan *et al* and Bai *et al* [9, 10] has indicated through simulation and calculation that Ti element improves the fracture toughness and ductility at room temperature of refractory alloys and RHEAs by decreasing P-N energy barrier ($U_{P\text{-}N}$) to overcome for dislocation movement. Moreover, Ti especially Cr and Al elements are considered as very beneficial high-temperature antioxidants due to the formation of protective TiO_2, Cr_2O_3 and Al_2O_3 oxide scale on the surface of some high melting point alloys [11–14]. Therefore, they are widely added to many high melting point alloys and RHEAs, even though the latter two elements degrade the fracture toughness

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and ductility of these alloys to some extent [15]. For this reason, MoNbHfCrTiAl system refractory high entropy alloy has been selected in this work.

As an active element with relatively higher melting point, Zr is also often added to some RHEAs [16, 17]. Yurchenko and Senkov *et al* [18, 19] have pointed out that Zr addition causes the formation of Zr and Al-rich intermetallic compounds and increases the compressive yield strength of RHEAs by solution and precipitation strengthening effects. Tong *et al* [20] has reported that Zr alloying improves both the strength and ductility of RHEAs by forming Zr-Zr metallic bonds. Ji *et al* [21] has developed a series of $Ti_{65-x}Ta_{25}Nb_{10}Zr_x$ RHEAs with single BCC phase, and indicated that Zr addition dramatically alleviates the atomic-size and modulus mismatches and therefore ameliorates the mechanical properties of the alloys. However, the study regarding roles of Zr on oxidation resistance of RHEAs and related mechanism has rarely been involved. Taking the optimization of composition for RHEAs into consideration, it is necessary to integrally evaluate Zr alloying effects on their microstructures, mechanical properties and oxidation resistance in different RHEA systems especially with more constituent elements. In the present work, three alloys with nominal compositions of $Mo_{0.5}NbHf_{0.25}CrZr_xTiAl$ (x = 0, 0.5, 1, mole ratio) have been designed to investigate the influences of Zr addition on the microstructure, compression performance and oxidation resistance of RHEAs at high temperature. For the convenience of description, the three RHEAs in this study are referred to 0Zr, 0.5Zr and 1Zr ones.

2. Experimental procedures

Three button ingots of RHEAs weighing approximately 200 g for each one were prepared using vacuum non-consumable arc melting technology, and they were remelted for six time in order to ensure compositional homogeneity. The purities of all raw materials are greater than 99.9 wt%. The specimens used for compressive test were cylinders of 6 mm in diameter and 9 mm in length, which were cut from the button ingots of RHEAs by electro-discharge machining (EDM). The compressive test was conducted at 1473 K with a strain rate of 0.005 s^{-1} and a height reduction of 0.35 using a Gleeble 3500 testing machine. The specimens were heated to target temperature at a temperature rising rate of 3 K s^{-1}, and then held for three minutes, prior to testing, in order to homogenize the temperature throughout the heating area.

The specimens used for oxidation tests were 8 mm × 8 mm × 8 mm cubes by EDM. The isothermal oxidation behavior of the cubic specimens was examined using an open-ended tube electric furnace in static air. After holding them at 1473 K for 4 or 50 h, the specimens were air-cooled down to room temperature. The mass gains per unit area of the specimens after oxidation were obtained by the formula $\Delta m\ A^{-1}$, where Δm is the mass change of the specimen before and after oxidation and A represents its starting surface area.

The constituent phases of arc melted, compressed and oxidized specimens were identified by x-ray diffractometer (XRD, Panalytical X'Pert Pro, Cu K_α). The microstructure of the specimens was observed by scanning electron microscope (SEM, QUANTA FEG 250 and TESCAN VEGA3 LMU) using backscattered electron (BSE) imaging mode, and the chemical compositions of their constituent phases were determined by energy dispersive spectrometer (EDS, INCA X-sight). In addition, the quantitative analysis of the constituent phases was carried out using Image-Pro Plus 6.0 high resolution imaging analysis software.

3. Results and discussion

3.1. Microstructure of RHEAs

Figures 1 and 2 show the XRD patterns and BSE images of the three RHEAs and table 1 presents the chemical compositions of their constituent phases. It can be seen from figure 1 that the three RHEAs all consist of BCC1, BCC2 and Laves phases (the former two phases have body-centered cubic structure and the latter one has Cr_2Nb type C14 structure), and the amount of diffraction peaks from BCC2 phase increase with Zr content in the alloy. Combined with XRD, BSE and EDS analysis results, the microstructures of both 0Zr and 0.5Zr RHEAs are primarily composed of thick BCC1 dendrites and BCC1 + BCC2 eutectics (BCC1 and BCC2 phases have dark and light contrasts respectively), as shown in figures 2(a)–(d). The average size of BCC1 dendrites is smaller while the area fraction of BCC1 + BCC2 eutectics is larger in the latter RHEA than in the former one (about 31.5% and 57.1% in 0Zr and 0.5Zr RHEAs respectively). With the further addition of Zr, the microstructure of 1Zr RHEA mainly comprises BCC1 + BCC2 eutectics (figure 2(e)–(f)). Moreover, there exists an increasing area fraction of BCC2 phase with increase in Zr content (about 20.4%, 37.9% and 52.3% in 0Zr, 0.5Zr and 1Zr HREAs, respectively). According to the result in table 1, BCC2 phase possesses obviously higher Hf, Cr, especially Zr concentrations than BCC1 one. In other words, Zr element is inclined to dissolve in BCC2 phase compared to BCC1 one. Moreover, the magnitude of increase in Zr concentration is significantly higher in

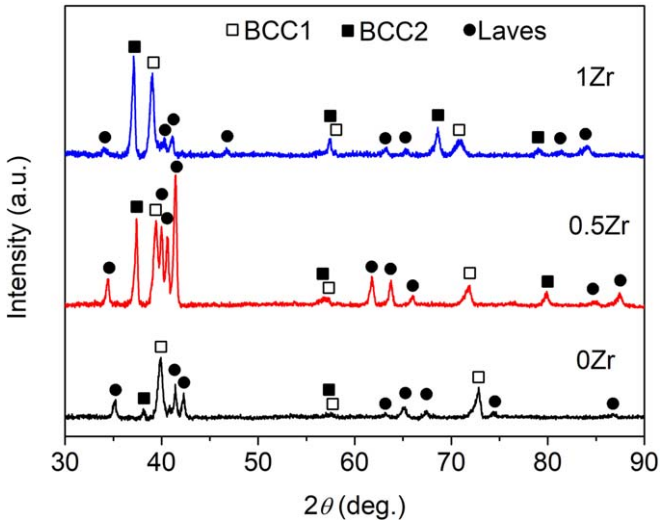


Figure 1. XRD patterns of the three RHEAs.

BCC2 phase than in BCC1 one with increase in Zr content in RHEA, which illustrates that Zr addition promotes the formation of BCC2 phase to some extent.

It should be noted that although the diffraction peaks from Laves phase are present in XRD patterns of 0Zr, 0.5Zr and 1Zr RHEAs (figure 1), this phase has not directly been observed in their BSE images. According to the open literatures [22], when parameter $\Omega \geqslant 1.1$ and atomic size difference parameter $\delta_r \leqslant 6.6\%$ ($\Omega = T_m\Delta S_{mix}/|\Delta H_{mix}|$, T_m, ΔS_{mix} and ΔH_{mix} represent effective melting temperature, mixing entropy and mixing enthalpy, respectively) are both true, only solid solution phase stabilizes in the high entropy alloys. Otherwise when $\Omega < 1.1$ or $\delta_r > 6.6\%$ is true, the intermetallic compounds would form in the high entropy alloys. Table 2 gives the calculated values of these several parameters of the three RHEAs in this study. It can be seen that in spite of $\Omega > 1.1$, $\delta_r > 6.6\%$ is true, indicating the formation of intermetallic compounds in 0Zr, 0.5Zr and 1Zr RHEAs, which is in agreement with the XRD result that the diffraction peaks from Cr_2Nb type Laves phase (an intermetallic compound) are present in XRD patterns of these three alloys. Furthermore, Senkov *et al* [23] has also reported that the Cr_2Nb type Laves phase exists in the form of extremely fine particles in BCC solid solution of $Ta_{0.5}Mo_{0.5}NbCrZrTi$ RHEA similar to $Mo_{0.5}NbHf_{0.25}CrZr_xTiAl$ ones designed in this paper, which is mainly attributed to the larger atomic size difference between Cr and Nb, Mo, Hf etc elements. It can be therefore inferred that the Laves phase could have been dispersively precipitated within BCC phases (especially in BCC2 phase with obviously higher Cr concentration than BCC1 one) during cooling after arc melting for RHEAs in this work.

3.2. Compressive behavior of RHEAs

Figure 3 shows the engineering stress-strain curves of the three RHEAs compressed at 1473 K with a strain rate of 0.005 s^{-1}. As can be seen, Zr addition obviously increases the compressive yield strength of RHEAs, and there exist two peak stresses, which are observed at the strains of near 0.03 and 0.08 respectively in the curves of 0Zr, 0.5Zr and 1Zr RHEAs. The former peak stress shows an increasing trend with Zr addition (about 160, 182 and 253 MPa for 0Zr, 0.5Zr and 1Zr HREAs respectively). The latter peak stress is significantly higher than the former peak stress for 0Zr and 0.5Zr HREAs, however, the two peak stresses are relatively close for 1Zr HREA (the latter peak stress are about 181, 255 and 246 MPa for 0Zr, 0.5Zr and 1Zr HREAs respectively).

Tang and Guo *et al* have [24] also found two peak stresses at lower and higher strains respectively in the compressive stress-strain curves of Nb-Si based ultrahigh temperature alloys composed of Nb based solid solution (Nbss) and Nb_5Si_3 intermetallic compound. And they have pointed out that the peak stresses at lower and higher strains are mainly caused by the deformation hardening of Nbss and Nb_5Si_3 phases, respectively. It is therefore suggested that the deformation hardening of the two BCC solid solutions and Cr_2Nb type Laves phases are probably responsible for the peak stresses at the strains of near 0.03 and 0.08 respectively.

Figure 4 shows the microstructures of the three RHEAs upon compression at 1473 K. It can be seen that compared with arc melted RHEAs, the morphologies of thick BCC1 dendrites and BCC1 + BCC2 eutectics are less visible in compressed ones and their microstructures become relatively uniform. Furthermore, the BCC1 and BCC2 phases especially in 0Zr and 0.5Zr RHEAs tend to deform in a direction perpendicular to the

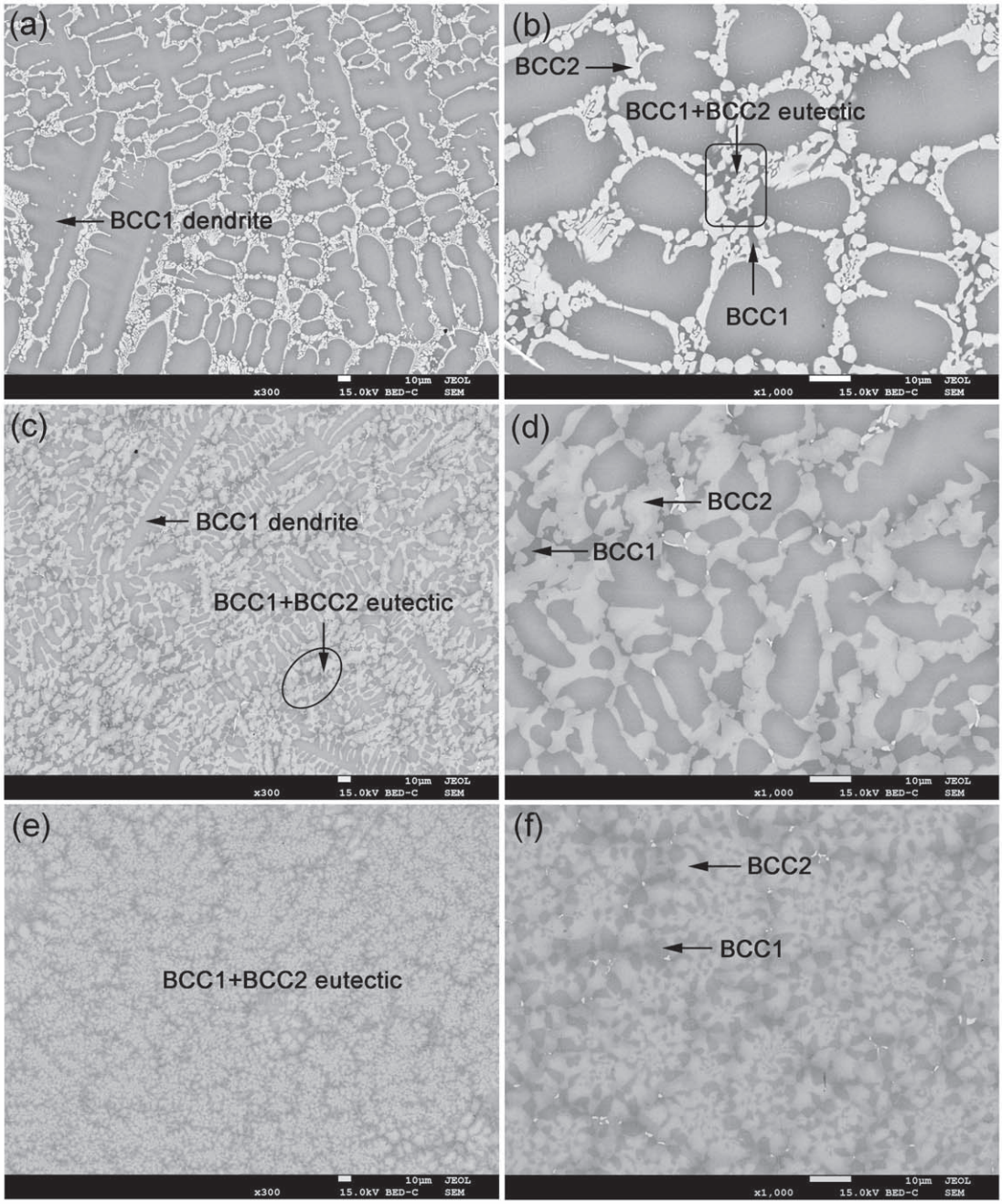


Figure 2. BSE images of the microstructures of the three RHEAs: 0Zr (a), (b); 0.5Zr (c), (d) and 1Zr (e), (f).

Table 1. Chemical compositions of the constituent phases in 0Zr, 0.5Zr and 1Zr RHEAs, determined by EDS.

Alloy	Phase	Chemical composition (at%)						
		Mo	Nb	Hf	Cr	Zr	Ti	Al
0Zr	BCC1$^{\#}$	15.1	25.4	3.4	16.3	/	20.8	19.0
	BCC1$^{\#\#}$	10.2	20.5	4.8	20.0	/	21.9	22.6
	BCC2$^{\#\#}$	5.2	16.3	13.0	33.5	/	10.3	21.8
0.5Zr	BCC1$^{\#}$	14.1	25.7	3.1	13.7	3.4	24.6	15.4
	BCC1$^{\#\#}$	9.6	23.7	2.0	11.5	2.8	33.4	17.0
	BCC2$^{\#\#}$	7.6	12.6	8.0	28.5	11.6	10.5	21.2
1Zr	BCC1$^{\#\#}$	12.1	28.9	3.0	10.3	6.8	26.9	12.0
	BCC2$^{\#\#}$	10.2	13.1	7.1	23.1	19.0	9.6	17.9

The superscripts '$^{\#}$' and '$^{\#\#}$' mean dendritic and eutectic phases respectively.

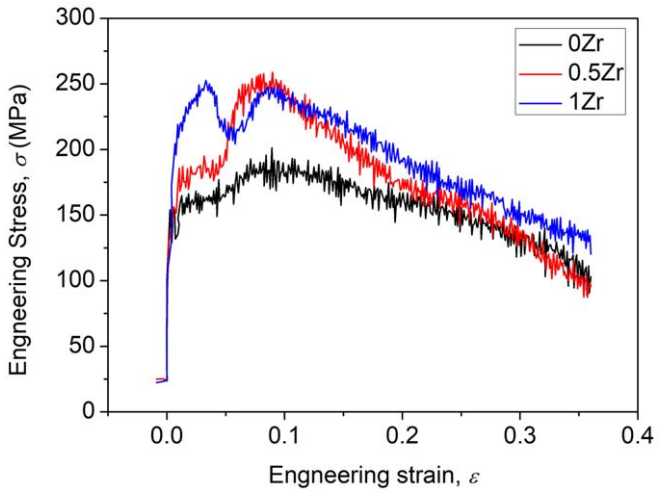


Figure 3. Engineering stress-strain curves of the three RHEAs compressed at 1473 K with a strain rate of 0.005 s^{-1}.

Table 2. Calculated values of parameters T_m, ΔS_{mix}, ΔH_{mix}, Ω and δ_r in 0Zr, 0.5Zr and 1Zr RHEAs.

Alloy	T_m (K)	ΔS_{mix} (J mol^{-1} K^{-1})	ΔH_{mix} (kJ mol^{-1})	Ω	δ_r (%)
0Zr	2064.40	14.17	−15.78	1.85	6.74
0.5Zr	2070.17	15.43	−16.91	1.89	7.30
1Zr	2074.94	15.55	−17.42	1.85	7.72

compression direction. It is worth noting that some tiny holes are observed in the microstructure of compressed RHEAs, as shown in figure 4, which should be ascribed to the stress concentration resulted during compression.

It can be seen from figure 1 that the most of diffraction peaks from the constituent phases shift to the left with increase in Zr content, indicating the increasing lattice parameters and lattice distortion effect (the lattice parameters of BCC1 and BCC2 phases are 0.3207 and 0.3338 nm, 0.3227 and 0.3399 nm, 0.3262 and 0.3423 nm in 0Zr, 0.5Zr, 1Zr RHEAs, respectively), which therefore brings about a great solution strengthening effect. As mentioned above, the amount of BCC2 phase with very high Cr concentration rises with increase in Zr content in RHEA, implying the formation of possibly more Cr_2Nb type Laves hard phase. Furthermore, according to our previous studies [25], BCC2 phase with more Hf, Cr and Zr elements has higher hardness than BCC1 one. All these illustrate the increased compression yield strength with Zr addition in RHEAs to some extent.

3.3. Oxidation behavior of RHEAs

The oxide scales of the all three RHEAs after oxidation at 1473 K for 4 and 50 h did not peel off their surfaces. It can be seen from figure 5 that the mass gains per unit area of 0Zr, 0.5Zr and 1Zr RHEAs after oxidation for 4 and 50 h, are 7.5, 35, 56.1 mg cm^{-2} as well as 77.5, 191.6, 208.8 mg cm^{-2} respectively, suggesting the gradually degraded oxidation resistance with Zr addition. This seems inconsistent with related reports such as that Zr addition improves the oxidation resistance of refractory Nb-Si based ultrahigh temperature alloys [18, 19]. The microstructures of 0Zr and 0.5Zr RHEAs after oxidation at 1473 K for 4 h have been observed and analyzed, aiming at interpretation of degradation effect of Zr on oxidation resistance.

Figures 6 and 7 show the BSE images of cross-sectional microstructures of the oxidized 0Zr and 0.5Zr RHEAs. It can be seen that the thickness of oxide scale of 0.5Zr RHEA is significantly higher than that of 0Zr one (about 281 and 112 μm respectively), which is consistent with the result regarding the mass gain after oxidation. There exists obviously layered structure for the oxide scales of the two RHEAs (namely outer, middle and inner layers of the oxide scale). And the thickness ratio of inner layer of the oxide scale is evidently higher in 0.5Zr RHEA than in 0Zr one (about 66% and 33% respectively, figures 6(a) and 7(a)). Notably, the compactness of the oxide scale of RHEA decreases with Zr addition (many cracks and holes are observed in the oxide scale of 0.5Zr RHEA, as shown in figure 7). In addition, an internal oxidation zone divided into both upper and lower parts is observed on surface of RHEA substrate, as shown in figures 6(a) and 7(a).

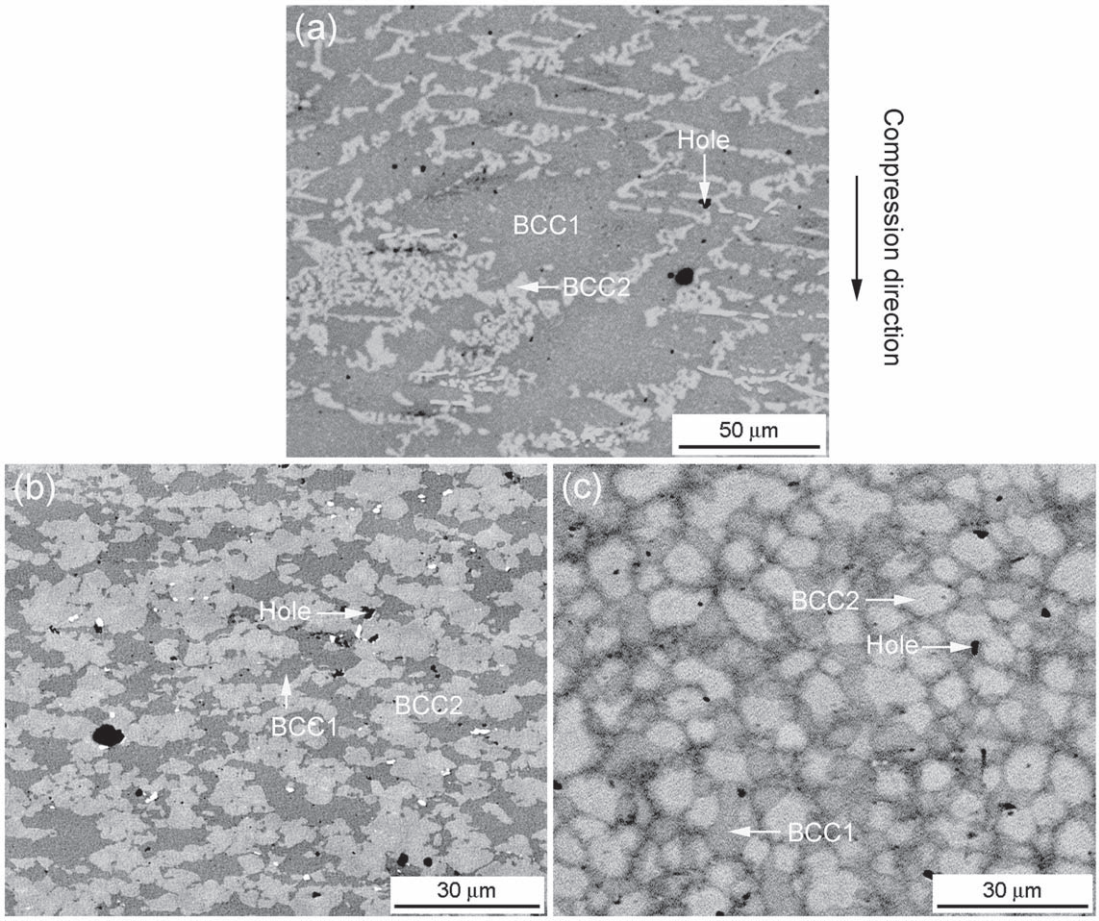


Figure 4. BSE images of the microstructures of the three RHEAs upon compression at 1473 K: 0Zr (a); 0.5Zr (b) and 1Zr (c).

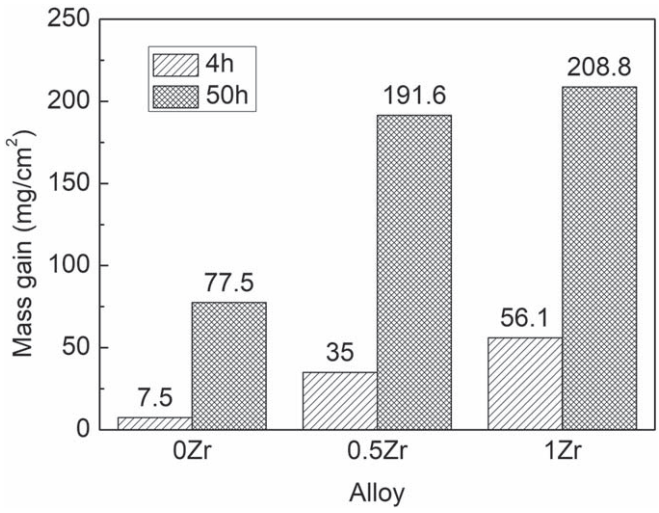


Figure 5. Mass gains per unit area of the three RHEAs upon oxidation at 1473 K for 4 and 50 h.

Figure 8 shows the XRD patterns of the oxide scales of 0Zr and 0.5Zr RHEAs. Table 3 gives the chemical compositions of oxides in oxide scales and internal oxidation zones of these two RHEAs. Combined with BSE, XRD and EDS analysis results, the out layer of oxide scale of 0Zr RHEA is mainly composed of black needle-like Al_2O_3 and light gray MoO_3 distributed on the surface. The oxides in the middle layer are primarily $(Ti,Cr,Nb)O_2$ (a solid solution oxide with TiO_2 crystal structure [26]) and $CrNbO_4$. These two oxides cannot be distinguished due to their similar gray contrasts (figures 6(b) and 8, table 3). In addition to the four oxides above, a solid

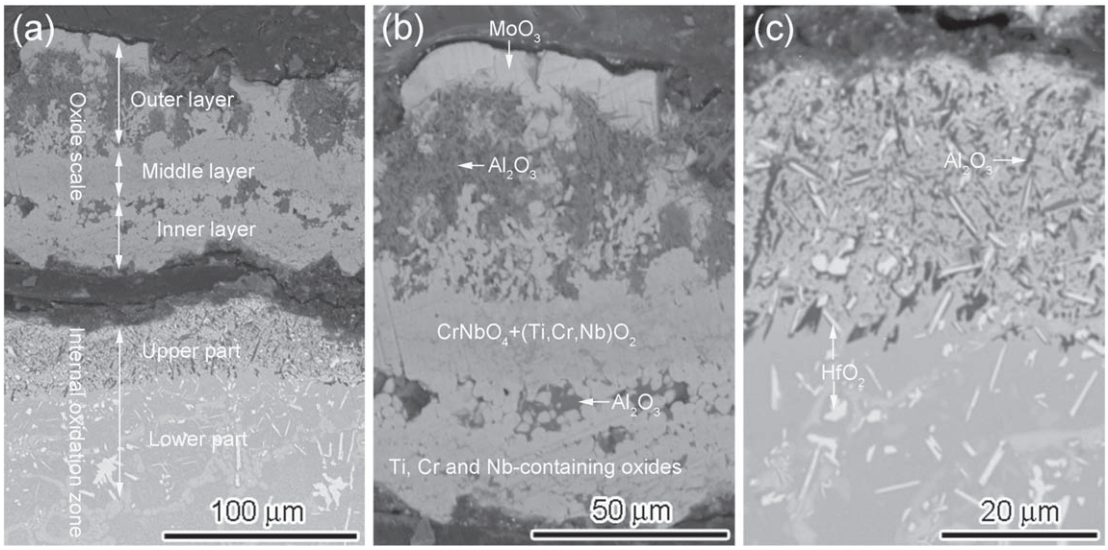


Figure 6. Typical cross-sectional BSE image of 0Zr RHEA upon oxidation at 1473 K for 4 h (a) and details of the outer, middle, inner layers of oxide scale (b) and internal oxidation zone (c).

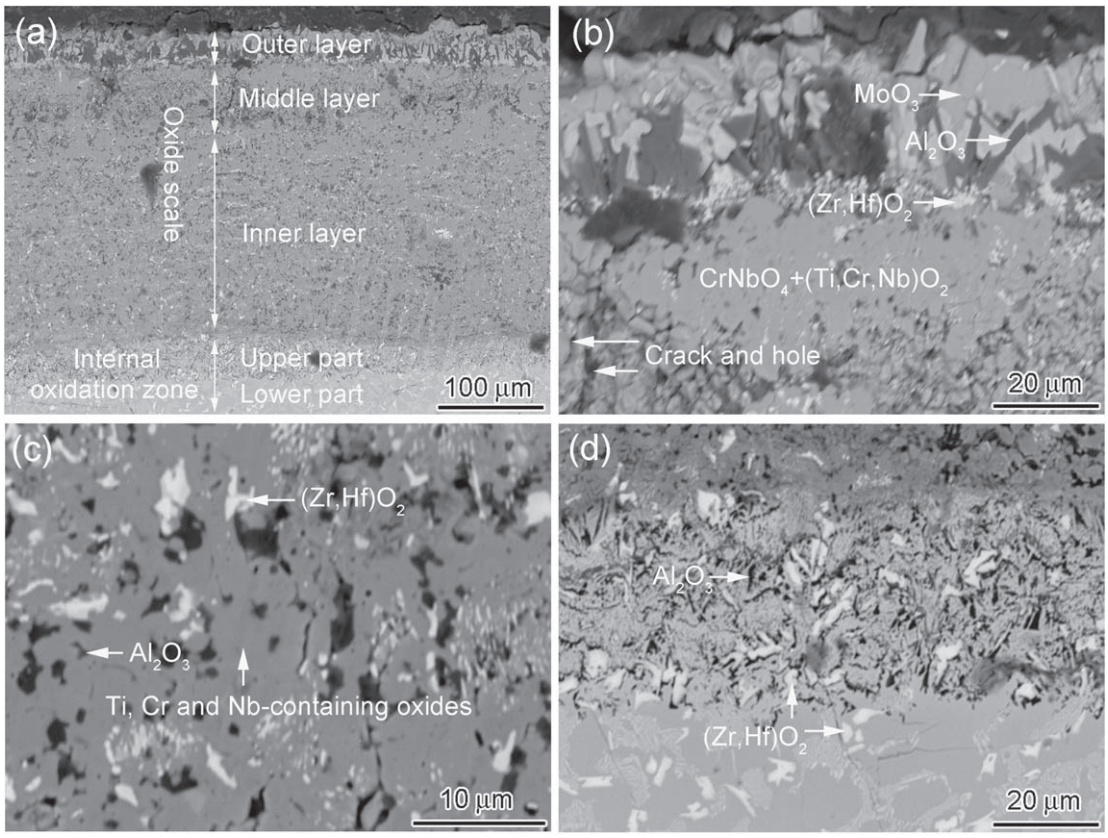


Figure 7. Typical cross-sectional BSE image of 0.5Zr RHEA upon oxidation at 1473 K for 4 h (a) and details of the outer, middle (b), inner (c) layers of oxide scale and internal oxidation zone (d).

solution oxide with ZrO_2 crystal structure, namely white granular $(Zr,Hf)O_2$ is also observed in the outer and middle layers of the oxide scale of 0.5Zr RHEA (figures 7(b) and 8, table 3).

Take the shallow depth of penetration in x-ray diffraction into consideration, the oxides in inner layer of the oxide scale and internal oxidation zone are difficult to be identified by XRD but EDS instead. The microstructure of inner layer of the oxide scale mainly consists of black Al_2O_3 and gray Ti, Cr and Nb-containing oxides for 0Zr RHEA, and besides oxides above, also white $(Zr, Hf)O_2$ for 0.5Zr one (figures 6(b), 7(c) and 8, table 3). It can be

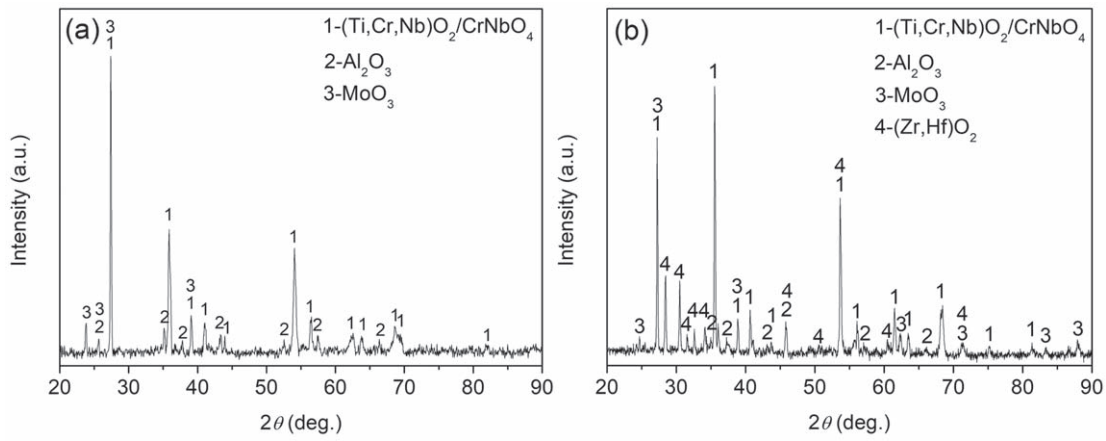


Figure 8. XRD patterns of the oxide scales formed on 0Zr (a) and 0.5Zr (b) RHEAs upon oxidation at 1473 K for 4 h.

Table 3. Chemical compositions of the oxides in the oxide scale and internal oxidation zone of 0Zr and 0.5Zr RHEAs after oxidation at 1473 K for 4 h, determined by EDS.

Alloy	Oxide	Chemical composition (at%)								
		Mo	Nb	Hf	Cr	Zr	Ti	Al	O	
0Zr	OL	MoO_3	21.22	/	/	0.18	/	/	0.11	78.49
		Al_2O_3	0.52	0.13	/	3.39	/	0.02	29.81	66.13
	ML	$CrNbO_4$+(Ti,Cr,Nb)O_2	0.63	9.12	0.61	7.42	/	9.06	1.28	71.88
	IL	Al_2O_3	1.67	2.36	0.77	0.33	/	1.18	18.89	74.82
		Ti, Cr and Nb-containing oxides	0.76	8.46	0.55	7.87	/	9.12	1.11	72.13
	IOZ	Al_2O_3	0.98	1.93	0.77	0.09	/	1.71	22.88	71.64
		HfO_2	0.83	1.91	22.76	0.06	/	2.11	1.32	71.01
0.5Zr	OL	MoO_3	22.08	0.11	/	0.11	0.08	0.02	2.23	75.37
		Al_2O_3	0.11	0.29	0.01	2.98	0.22	/	26.55	69.84
	ML	$CrNbO_4$+(Ti,Cr,Nb)O_2	0.45	8.56	0.73	7.94	1.21	8.89	1.09	71.13
		(Zr,Hf)O_2	/	1.45	9.12	0.31	15.88	0.22	1.01	72.01
	IL	Al_2O_3	0.32	1.11	/	1.23	0.12	0.89	31.12	65.21
		Ti, Cr and Nb-containing oxides	0.67	9.19	0.75	9.88	1.11	8.14	1.78	68.48
		(Zr,Hf)O_2	0.38	1.37	8.76	0.61	15.73	1.19	/	71.96
	IOZ	Al_2O_3	0.21	0.45	/	1.32	0.43	0.45	28.91	68.23
		(Zr,Hf)O_2	0.22	1.22	8.93	0.36	16.26	0.18	0.89	71.94

'OL', 'ML', 'IL' and 'IOZ' represent the outer layer, middle layer, inner layer of oxide scale and internal oxidation zone of RHEAs after oxidation at 1473 K.

reasonably deduced that Ti, Cr and Nb-containing oxides might be mixture of TiO_2, Cr_2O_3, Nb_2O_5, $CrNbO_4$ and (Ti,Cr,Nb)O_2. Attentively, Hf oxide, i.e. HfO_2 is not determined and Hf tends to be dissolved in other oxides in the oxide scale of 0Zr RHEA, while (Zr,Hf)O_2 as the main existence form of Hf element, is present in that of 0.5Zr one by BSE, XRD and EDS analyses (discussed below). The results about identification of oxides in the oxide scale are basically confirmed by the typical EDS mapping images from 0Zr RHEA, as shown in figure 9.

The oxides in the upper part of internal oxidation zone are mainly Al_2O_3 and HfO_2 for 0Zr RHEA, as well as Al_2O_3 and (Zr,Hf)O_2 for 0.5Zr one (Al_2O_3 is black particle, HfO_2 or (Zr, Hf)O_2 is white one). However, only white HfO_2 or (Zr, Hf)O_2 particle is detected in the lower part of internal oxidation zone for 0Zr or 0.5Zr RHEA respectively (figures 6(c), 7(d) and 8, table 3).

According to the distribution of oxides in the oxide scale and internal oxidation zone, it can be concluded that the initial oxides form at the interface area between oxide scale and RHEA substrate, and thus the oxidation process of RHEAs is mainly dominated by inward diffusion of oxygen element in comparison with outward diffusion of metal elements. At high temperatures, oxygen absorbs and dissolves on the surface of RHEA, and then diffuses into its matrix. When the oxygen is saturated in RHEA matrix, some fine oxides such as HfO_2 and ZrO_2 from reactive elements Hf and Zr, preferentially precipitate in it (table 4 presents the standard free energies of formation of oxides from Mo, Nb, Hf, Cr, Zr, Ti and Al elements at 1473 K). Moreover, HfO_2 would be further dissolved in ZrO_2 to form (Zr,Hf)O_2 in 0.5Zr RHEA by BSE, XRD and EDS analyses, implying that

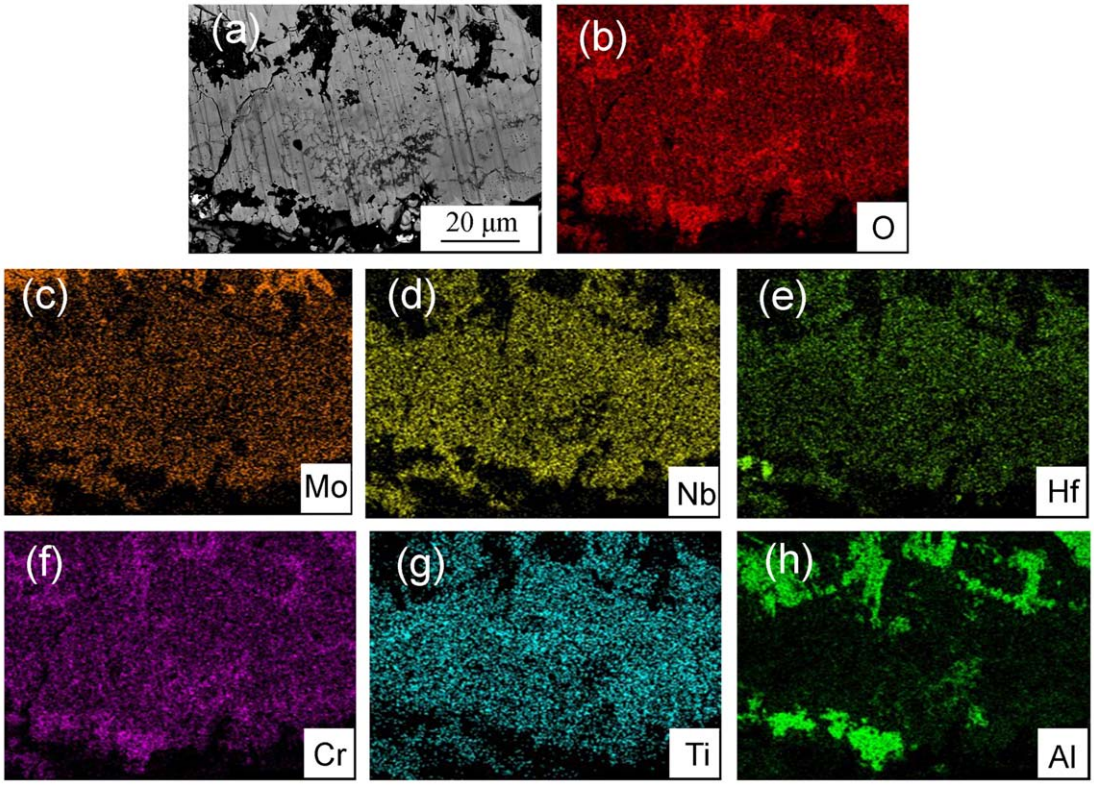


Figure 9. Typical EDS mapping images of the oxide scale of 0Zr RHEA upon oxidation at 1473 K for 4 h.

Table 4. Standard free energies of formation of oxides at 1473 K.

Oxide	MoO_3	Nb_2O_5	HfO_2	Cr_2O_3	ZrO_2	TiO_2	Al_2O_3
ΔG^{θ} (kJ mol^{-1})	268	507	850	502	822	679	803

higher stability of latter oxide than the former two ones. Subsequently, Al element starts to react with oxygen accompanied by the precipitation of Al_2O_3. As thus, the internal oxidation zone forms at first.

With increase in oxygen concentration, Hf, Zr and Al elements with higher affinities for oxygen have been gradually consumed, and also the oxidation of Ti, Nb and Cr elements occur to form TiO_2, Nb_2O_5 and Cr_2O_3. Combined with the microstructure of oxide scale in this paper and related results in open literatures [8, 25], it is reasonably inferred that the three oxides above would react and dissolve with each other to form $CrNbO_4$ and $(Ti,Cr,Nb)O_2$, and all of these correspond to the mixture of Ti, Cr and Nb-containing oxides in the inner layer of oxide scale. Also, HfO_2 tends to be dissolved in the oxides already formed in 0Zr RHEA while $(Zr,Hf)O_2$ still steadily exists in 0.5Zr one by BSE, XRD and EDS analyses. Because of its higher stability, Al_2O_3 is difficult to react with other oxides to form more complex oxidation products.

With further inward diffusion of oxygen, the mixture of Ti, Cr and Nb-containing oxides would almost completely convert to $CrNbO_4$ and $(Ti,Cr,Nb)O_2$, and also Mo element has gradually been oxidized to MoO_3 due to high enough oxygen concentration (MoO_3 has the lowest standard free energies of formation in this work). It is worth mentioning that MoO_3 is volatile at elevated temperature and therefore accumulates at the surface of oxide scale. Meanwhile, the original Al_2O_3 would be pushed off $CrNbO_4$ and $(Ti,Cr,Nb)O_2$, and move towards MoO_3, thus leading to the formation of middle and outer layers of the oxide scale. Also, the fine and stable $(Zr,Hf)O_2$ is still distributed at middle $CrNbO_4 + (Ti,Cr,Nb)O_2$ layer and outer $Al_2O_3 + MoO_3$ one.

As mentioned above, compared to HfO_2, $(Zr,Hf)O_2$ is distributed in the whole oxide scale and internal oxidation zone due to its high stability. The existence of $(Zr,Hf)O_2$ would make the oxide scale brittle to some extent. Therefore, the cracks and holes are easily formed under high stress induced by the growth of oxides around $(Zr,Hf)O_2$ in the oxide scale (figure 7), which is importantly responsible for the decreased compactness of the oxide scale of 0.5Zr RHEA compared to 0Zr one. The considerable cracks and holes would become channels for rapid inward diffusion of oxygen element in the oxide scale, thus resulting in degraded oxidation resistance of Zr-containing RHEAs.

4. Conclusions

(1) The three RHEAs all comprise BCC1, BCC2 and Laves phases. The microstructures of 0Zr and 0.5Zr RHEAs are mainly composed of thick BCC1 dendrites and BCC1 + BCC2 eutectics, while that of 1Zr one primarily of BCC1 + BCC2 eutectics. With increasing the Zr content, the area fraction of BCC2 phase gradually increases (20.4%, 37.9% and 52.3% for 0Zr, 0.5Zr and 1 Zr RHEAs respectively).

(2) There exist two peak stresses observed at the strains of near 0.03 and 0.08 respectively in the engineering stress-strain curves of the three RHEAs compressed at 1473 K. Generally, the compressive yield strength of RHEA exhibits an increasing trend with Zr addition.

(3) All the oxide scales of the three RHEAs upon oxidation at 1473 K possess good adhesion and significantly layered structures. And also, an internal oxidation zone is observed between RHEA substrate and oxide scale. The oxidation products dramatically vary in the oxide scale and internal oxidation zone. The oxidation resistance of RHEA is obviously degraded due to the decreased compactness of the oxide scale with Zr addition based on the formation of $(Zr,Hf)O_2$.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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